

NEWINGTON COLLEGE



2014

Preliminary examination Mathematics Extension 1

General Instructions:

- Date of task - Wednesday 22nd August
- Working time – 120 mins
- Board-approved calculators may be used.
- Attempt all questions, start each question in a new booklet.
- Show all relevant mathematical reasoning and/or calculations.

Total marks – 70

Outcome	Marks
Section I – Multiple choice	/10
Section II – Q11 (Trigonometry & Functions)	/12
Section II – Q12 (Geometry of the derivative)	/12
Section II – Q13 (Sequences and series)	/12
Section II – Q14 (Induction & Quadratic functions)	/12
Total	/70

Section I (10 marks)**Allow about 15 minutes for this section****Use the multiple-choice answer sheet for Questions 1-10**

Question 1The co-ordinates of the focus of the parabola $x^2 = 4ay$ are

- (a) $(0, -a)$ (b) $(0, a)$ (c) $(0, 1)$ (d) $(0, 4a)$

Question 2If α and β are the zeroes of the quadratic polynomial $y = 3x^2 + x - 9$ then

- (a) $\alpha + \beta = -\frac{1}{3}$ (b) $\alpha + \beta = \frac{1}{3}$ (c) $\alpha + \beta = -3$ (d) $\alpha + \beta = 3$

Question 3The 20th term of the arithmetic series $4 + 5\frac{1}{3} + 6\frac{2}{3}$ is

- (a) $4 + (20 - 1) \times 1\frac{1}{3}$ (b) $\frac{20}{2} \left[2 \times 4 + (20 - 1) \times 1\frac{1}{3} \right]$ (c) $4 + 20 \times 1\frac{1}{3}$ (d) $4 \times \left(\frac{4}{3} \right)^{20-1}$

Question 4 $\sum_{r=1}^{r=5} (-3)^r$ is equal to

- (a) 45 (b) -183 (c) -182 (d) -363

Question 5Given $f(x) = 3x^2 - 5x + 2$, find $f(\alpha + 1)$

- (a) $3\alpha^2 - 5\alpha + 3$ (b) $3\alpha^2 + 11\alpha$ (c) $3\alpha^2 + \alpha + 1$ (d) $3\alpha^2 + \alpha$

Question 6

The perpendicular distance, in units, from $(1,0)$ to $3x - 4y = 6$ is

- (a) $\frac{9}{5}$ (b) $\frac{3}{5}$ (c) $\frac{7}{5}$ (d) $-\frac{9}{5}$

Question 7

If $y = (3x-1)\sqrt{3x-1}$ then $\frac{dy}{dx}$ is

- (a) $2\sqrt{3x-1}$ (b) $\frac{9}{2\sqrt{3x-1}}$ (c) $\frac{9}{2}\sqrt{3x-1}$ (d) $3\sqrt{3x-1} - \frac{3}{\sqrt{3x-1}}$

Question 8

The natural domain of the function $y = \sqrt{5-x^2}$ is

- (a) $-5 < x < 5$ (b) $-5 \leq x \leq 5$ (c) $-\sqrt{5} \leq x \leq \sqrt{5}$ (d) $0 < y < \sqrt{5}$

Question 9

The equation of the line perpendicular to $3x + 4y - 3 = 0$, which also passes through the point $(-1, -4)$ is

- (a) $4x - 3y + 13 = 0$ (b) $3x - 4y - 13 = 0$ (c) $4x + 3y + 16 = 0$ (d) $4x - 3y - 8 = 0$

Question 10

The expression $\frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta}$ may be simplified to

- (a) $2\sec^2\theta$ (b) $2\sin^2x$ (c) $2\cos^2x$ (d) $2\operatorname{cosec}^2x$

Section II (60 marks)**Attempt Questions 11 – 15****Allow about 1 hours and 45 minutes for this section****Question 11 (12 marks) – Start a new booklet****Marks**

- (a) Prove $(\cot \theta + \operatorname{cosec} \theta)^2 = \frac{1 + \cos \theta}{1 - \cos \theta}$ **3**
- (b) Solve for $0 \leq \theta \leq 360^\circ$:
- (i) $2 \cos^2 \theta - \cos \theta = 0$, $0^\circ \leq \theta \leq 360^\circ$ **2**
- (ii) $5 \sec^2 \theta - 9 \tan \theta = 7$, $90^\circ \leq \theta \leq 180^\circ$. Give your answer correct to the nearest minute **3**
- (c)
- (i) Show that $y = \frac{2x}{x^2 + 4}$ is an odd function. **1**
- (ii) Find any intercepts. **1**
- (iii) Find the **equation** of the horizontal asymptote. **1**
- (iv) Sketch the curve. **1**

Question 12 (12 marks) – Start a new booklet

- (a) Differentiate with respect to x :
- (i) $y = \frac{1}{x^2} - 4\sqrt{2x-3}$ **2**
- (ii) $y = \frac{x^2-1}{x^2+1}$ Simplify your answer as much as possible. **2**
- (b) Find the co-ordinates of the points on the curve $y = 2x^3 - 9x^2 + 27$ where the tangent is horizontal. **3**
- (c) Find the value of the constant a , if the normal to the curve $y = \frac{a}{x+2}$ at the point where $x = 2$ has a gradient of -4 . **2**
- (d) If $f(x) = (x-4)^2 \times \sqrt[3]{2x+1}$, evaluate $f'(13)$. **3**

Marks**Question 13 (12 marks) – Start a new booklet**

- (a) A ball is dropped to the ground from a height of 2 metres. After each bounce it rebounds to half the height from which it fell.
- (i) Find the total distance it has travelled when it strikes the ground for the 4th time. **2**
- (ii) Using the formula for the sum of an infinite geometric progression, how far will it travel before it comes to rest? **2**
- (b) The 21st term of an arithmetic series is zero. The sum of the first 25 terms is 100. Find the first term and the common difference. **4**
- (c) The series $1+2+4+7+8+10+11+13+14+16+\dots+398$ was formed by omitting from the first 400 positive integers all those which are multiples of 3 or 5. Find the sum of the series. **4**

Question 14 (12 marks) – Start a new booklet

- (a) **Prove by mathematical induction** that $1+x+x^2+x^3+\dots+x^{n-1}=\frac{1-x^n}{1-x}$. **4**
- (b) Solve the equation $(x^2+x)^2=5x^2+5x-6$, giving your answers in **exact** form. **4**
- (c) If α and β are the roots of the equation $ax^2+bx+c=0$, express $(\alpha-2\beta)(2\alpha-\beta)$ in terms of a , b , and c . Hence deduce the condition for one root to be double the other. **4**

Marks

Question 15 (12 marks) – Start a new booklet

- (a) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.
- (i) Show that the equation of the tangent to the parabola at P is $y = px - ap^2$. 3
- (ii) The tangent at P and the tangent at Q intersect at T . Find the coordinates of T . 2
- (iii) If T is a point on the directrix of the parabola $x^2 = 4ay$, show that PQ is a focal chord. 2
- (c) $P(2t, t^2)$ is a point on the parabola $x^2 = 4y$ with focus S . The point M divides the interval SP externally in the ratio $3 : 1$.
- (i) Show that the co-ordinates of M are $\left(3t, \frac{3t^2 - 1}{2}\right)$ 3
- (ii) Show that as P moves on the parabola $x^2 = 4y$, the locus of M is given by the parabola $x^2 = 6y + 3$. 2

END OF EXAMINATION

Section I – Multiple choice answer sheet

10 Marks

Attempt Question 1 – 10.

Allow approximately 15 minutes for this section.

Use the multiple choice answer sheet below to record your answers to Question 1 – 10.

Select the alternative: A, B, C or D that best answers the question.

Colour in the response oval completely.

Sample:

$2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, draw a cross through the incorrect answer and colour in the new answer

ie A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word “correct” and draw an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct

Year 11 Extension 1 Mathematics

Multiple Choice Answer Sheet

Student number:

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Completely colour in the response oval representing the most correct answer.

1	A	☐	B	☐	C	☐	D	☐
2	A	☐	B	☐	C	☐	D	☐
3	A	☐	B	☐	C	☐	D	☐
4	A	☐	B	☐	C	☐	D	☐
5	A	☐	B	☐	C	☐	D	☐
6	A	☐	B	☐	C	☐	D	☐
7	A	☐	B	☐	C	☐	D	☐
8	A	☐	B	☐	C	☐	D	☐
9	A	☐	B	☐	C	☐	D	☐
10	A	☐	B	☐	C	☐	D	☐

Mark: /10

Solutions

Section I

1. B
2. A
3. A
4. B
5. D
6. B
7. C
8. C
9. D
10. A

Section II

Question 11

(a) $LHS = (\cot \theta + \operatorname{cosec} \theta)^2$

$$\begin{aligned} &= \left(\frac{\cos \theta + 1}{\sin \theta} \right)^2 \\ &= \frac{(\cos \theta + 1)^2}{\sin^2 \theta} \\ &= \frac{(\cos \theta + 1)^2}{1 - \cos^2 \theta} \\ &= \frac{(\cos \theta + 1)^2}{(1 - \cos \theta)(1 + \cos \theta)} \\ &= \frac{1 + \cos \theta}{1 - \cos \theta} = RHS \end{aligned}$$

(b) (i) $\cos \theta (2 \cos \theta - 1) = 0$

$$\cos \theta = 0 \text{ or } \cos \theta = \frac{1}{2}$$

$$\theta = 90^\circ, 270^\circ, 60^\circ \text{ or } 300^\circ.$$

(ii) $5(1 + \tan^2 \theta) - 9 \tan \theta = 7$

Let $x = \tan \theta$

$$5 + 5x^2 - 9x = 7$$

$$5x^2 - 9x - 2 = 0$$

$$(5x+1)(x-2) = 0$$

$$x = -\frac{1}{5} \text{ or } 2$$

$$\tan \theta = -\frac{1}{5} \text{ or } 2. \text{ But } 90^\circ \leq \theta \leq 180^\circ.$$

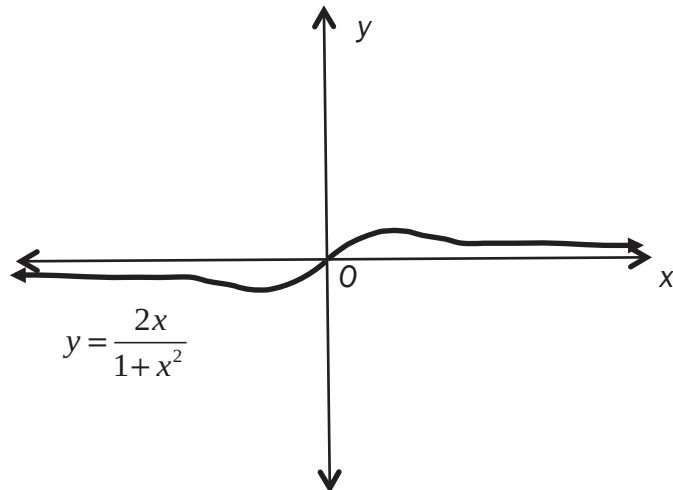
$$\theta = 168^\circ 41'$$

(c) (i)
$$f(-a) = \frac{-2a}{(-a)^2 + 4}$$
$$= \frac{-2a}{a^2 + 4}$$
$$= -f(a)$$

(ii) $(0,0)$

(iii) $y = 0$

(iv)



Question 12

(a) (i)
$$y' = \frac{-2}{x^3} - 4 \times \frac{1}{2} (2x-3)^{-\frac{1}{2}} \times 2$$
$$= \frac{-2}{x^3} - \frac{4}{\sqrt{2x-3}}$$

(ii)
$$y' = \frac{(x^2+1)2x - (x^2-1)2x}{(x^2+1)^2}$$
$$= \frac{4x}{(x^2+1)^2}$$

(b) $y = 2x^3 - 9x^2 + 27$

$$y' = 6x^2 - 18x = 0$$

$$6x(x-3) = 0$$

$$x = 0, x = 3$$

Hence, co-ordinates are (0,27) or (3,0)

$$(c) \quad y = \frac{a}{x+2}$$

Gradient of normal = -4. Hence, gradient of tangent is $\frac{1}{4}$.

$$y' = \frac{-a}{(x+2)^2} = \frac{1}{4} \text{ when } x = 2$$

$$\frac{-a}{(2+2)^2} = \frac{1}{4}$$

$$a = -4$$

$$(d) \quad f(x) = (x-4)^2 \times \sqrt[3]{2x+1} = (x-4)^2 \times (2x+1)^{\frac{1}{3}}$$

$$f'(x) = (x-4)^2 \times \frac{2}{3}(2x+1)^{-\frac{2}{3}} + (2x+1)^{\frac{1}{3}} \times 2(x-4)$$

$$f'(13) = (9)^2 \times \frac{2}{3}(27)^{-\frac{2}{3}} + (27)^{\frac{1}{3}} \times 2(9) \\ = 60$$

Question 13

$$(a) (i) \quad 2 + 1 + 1 + 0.5 + 0.5 + 0.25 + 0.25 = 5.5 \text{ metres}$$

$$(ii) \quad = 2 + 2 \left(1 + \frac{1}{2} + \frac{1}{4} + \dots \right) \\ = 2 + 2 \left(\frac{a}{1-r} \right), \text{ where } a = 1, r = \frac{1}{2} \\ = 2 + 2 \left(\frac{1}{1-\frac{1}{2}} \right) = 6 \text{ metres}$$

(b) Solve simultaneously:

$$a + 20d = 0 \dots\dots\dots 1 \quad \text{where } a = 1^{\text{st}} \text{ term and } d = \text{common difference}$$

$$\frac{25}{2}(2a + 24d) = 100 \dots\dots 2$$

$$a = 10, d = -\frac{1}{2}$$

$$(c) \quad 1+2+4+7+8+11+13+14+16+\dots+398$$

= sum of 1st 400 positive integers - multiples of 3 - multiples of 5 + multiples of 15

$$= \frac{400}{2} [2 \times 1 + (400-1)1] - \frac{133}{2} [2 \times 3 + (133-1)3] - \frac{80}{2} [2 \times 5 + (80-1)5] + \frac{26}{2} [2 \times 15 + (26-1)15]$$

$$\begin{aligned}
&= 200 \times 401 - 133 \times 201 - 40 \times 405 + 13 \times 405 \\
&= 42532
\end{aligned}$$

Question 14

(a) Prove $1 + x + x^2 + x^3 + \dots + x^{n-1} = \frac{1-x^n}{1-x}$

Step 1 - Prove result true for $n = 1$

$$\text{LHS} = x^{1-1} = 1$$

$$\text{RHS} = \frac{1-x^1}{1-x} = 1$$

$$\text{LHS} = \text{RHS}$$

Step 2 - Assume result true for $n = k$, where k is a positive integer.

$$\text{i.e. } 1 + x + x^2 + x^3 + \dots + x^{k-1} = \frac{1-x^k}{1-x}$$

Step 3 - Prove result true for $n = k + 1$.

$$\text{i.e. Prove } 1 + x + x^2 + x^3 + \dots + x^{k+1-1} = \frac{1-x^{k+1}}{1-x}$$

$$\text{Now, LHS} = 1 + x + x^2 + x^3 + \dots + x^{k+1-1}$$

$$= 1 + x + x^2 + x^3 + \dots + x^{k-1} + x^k$$

$$= \frac{1-x^k}{1-x} + x^k, \text{ from step 2.}$$

$$= \frac{1-x^k + x^k(1-x)}{1-x}$$

$$= \frac{1-x^k + x^k(1-x)}{1-x}$$

$$= \frac{1-x^k + x^k - x^{k+1}}{1-x}$$

$$= \frac{1-x^{k+1}}{1-x}$$

$$= \text{RHS}$$

Step 4 - By the principle of mathematical induction, the result has been proved true for all positive integers n .

(b) Let $y = x^2 + x$

Hence, $y^2 = 5y - 6$

$$\therefore y^2 - 5y + 6 = 0$$

$$(y-2)(y-3) = 0$$

$$y = 2 \text{ or } y = 3$$

$$\therefore x^2 + x = 2 \text{ or } x^2 + x = 3$$

$$\therefore x^2 + x - 2 = 0 \text{ or } x^2 + x - 3 = 0$$

$$(x+2)(x-1) = 0 \text{ or } x = \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times (-3)}}{2}$$

$$\therefore x = -2, 1, \frac{-1 \pm \sqrt{13}}{2}$$

(c) $ax^2 + bx + c = 0$

$$\alpha + \beta = \frac{-b}{a} \quad \alpha\beta = \frac{c}{a}$$

$$(\alpha - 2\beta)(2\alpha - \beta) = 2\alpha^2 - 5\alpha\beta + 2\beta^2$$

$$= 2(\alpha^2 + \beta^2) - 5\alpha\beta$$

$$= 2(\alpha^2 + 2\alpha\beta + \beta^2) - 9\alpha\beta$$

$$= 2(\alpha + \beta)^2 - 9\alpha\beta$$

$$= 2\left(\frac{-b}{a}\right)^2 - 9 \cdot \frac{c}{a}$$

$$= \frac{2b^2 - 9ac}{a^2}$$

Now, if $\alpha = 2\beta$ or $\beta = 2\alpha$ then $(\alpha - 2\beta)(2\alpha - \beta) = 0$

Hence $2b^2 = 9ac$

Question 15

(a) (i) $x^2 = 4ay$

$$2x = 4a \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x}{2a} \text{ at } x = 2ap$$

$$\frac{dy}{dx} = p$$

The equation of the tangent at p is:

$$y - ap^2 = p(x - 2ap)$$

$$= px - 2ap^2$$

$$\therefore y = px - ap^2$$

(ii) Tangent at P : $\therefore y = px - ap^2$ 1

Tangent at Q : $\therefore y = qx - aq^2$ 2

Sub 1 into 2

$$px - ap^2 = qx - aq^2$$

$$x(p - q) = ap^2 - aq^2$$

$$x = \frac{a(p^2 - q^2)}{p - q} = a(p + q)$$

Sub into 1.

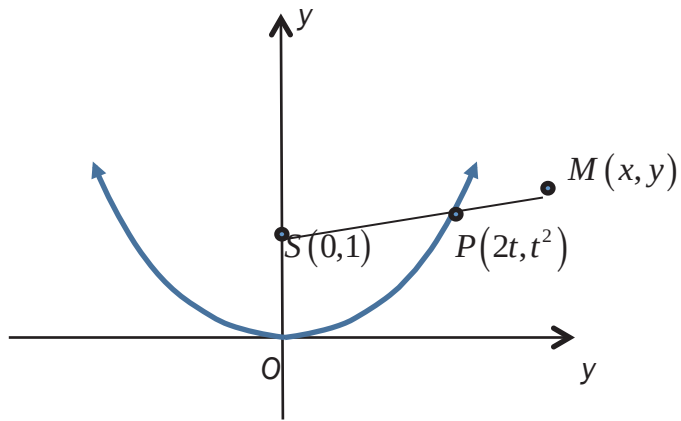
$$y = p.a(p + q) - ap^2 = apq$$

Hence, $T(a(p + q), apq)$

(iii) T on directrix $\rightarrow apq = -a$. Hence, $pq = -1$.

But p is the gradient of PT and q is the gradient of QT . Hence the tangents are perpendicular. Hence, $\angle PTQ = 90^\circ$.

(b) (i)



(i) 3:1 externally

$$x = \frac{3 \times 2t - 1 \times 0}{3 - 1} = 3t$$

$$y = \frac{3 \times t^2 - 1 \times 1}{3 - 1} = \frac{3t^2 - 1}{2}$$

Coordinates are $\left(3t, \frac{3t^2 - 1}{2} \right)$

(ii) $t = \frac{x}{3}$. Hence, $y = \frac{3\left(\frac{x}{3}\right)^2 - 1}{2}$

$$2y = \frac{x^2}{3} - 1$$

$$x^2 = 6y + 3$$